

Prediction Market Data Pipeline

Kalshi and Polymarket ingestion, cross-platform analysis, and research synthesis for a weekly internship deliverable

Executive summary

A pipeline polls Kalshi and Polymarket into Supabase on a schedule, paired with research into how professional traders think about pricing and resolution risk. 19 markets across crypto, macro and politics run on an unattended GitHub Actions cron, nominally every 2 hours. From the collected data I built a repricing-speed metric, compared volume, and measured divergence across 3 linked pairs. One early catch: a question-polarity mismatch that would have reported a false 80-percentage-point divergence on the clearest pair in the set.

The figures below cover 2026-09-02 to 2026-09-11, 849 snapshot rows across 19 markets. Cross-platform comparisons start 2026-09-04, since Polymarket ingestion came online after Kalshi's. The window is pinned in **analysis/config.py** and every number here is generated from it, so re-running reproduces the same report until the window is deliberately moved.

1. Architecture and pipeline

1.1 Schema

Six tables in Supabase (Postgres), with Row Level Security on. Writes use the `service_role` key, which bypasses RLS, so no read policies were needed at this stage.

Table	Purpose
platforms	Kalshi / Polymarket
markets	One row per tracked market, upserted on (platform_id, external_id)
market_snapshots	The core time-series table: one row per market per poll
resolutions	Final outcomes, written by the backfill step once a market settles
cross_platform_links	Manually matched market pairs tracking the same real-world event
kol_theses	Research log of prediction-market researcher theses and frameworks

1.2 Ingestion

Both platforms serve prices without authentication; only trading needs auth. Kalshi's **/trade-api/v2/markets/{ticker}** and Polymarket's **/markets/keyset** each return price plus metadata in one call, so snapshot polling needs no separate candlestick or order-book request.

The watchlist (12 Kalshi tickers and 7 Polymarket slugs) is fixed rather than a rolling top-N by volume, since markets dropping in and out would break the time series. One quirk: Kalshi gives every market in an event the same title, so all five Bitcoin strike buckets arrive as "BTC price on Jan 1, 2027?". The strike

lives in **yes_sub_title**, and without appending it the volume chart drew five identically labeled bars.

1.3 Scheduling and reliability

A workflow at the repo root (**.github/workflows/prediction-markets-w1.yml**, since Actions ignores workflows nested in subfolders) asks for both ingestion scripts every 2 hours, offset off the hour to dodge :00 queueing. A shared logger writes to the console and to **logs/pipeline.log**, which the workflow uploads as an artifact so it outlives the runner.

Requested and delivered are not the same thing. Across 46 poll batches over 229 hours the median gap is 4.6 hours rather than 2, the worst was 34.6 hours, and the rate works out to 4.8 batches a day against the 12 requested, about 40%. That looks like GitHub dropping scheduled events under load rather than anything specific to this repository.

Two things qualify that number. 40% overstates GitHub's share, because the batch count also includes manual **workflow_dispatch** runs and local runs, and rows don't record their trigger (Section 6). The batches are uneven too, with 42 of 46 writing all 19 markets and the other 4 listed below: three from the bring-up period before both scripts ran together, and one local Polymarket-only run.

- 2026-09-02 07:58 UTC: 12 markets, kalshi only
- 2026-09-03 18:35 UTC: 12 markets, kalshi only
- 2026-09-03 20:09 UTC: 7 markets, polymarket only
- 2026-09-09 19:37 UTC: 7 markets, polymarket only

A backfill step after each run checks every tracked market and records outcomes once they settle. The resolutions table holds 0 rows because nothing tracked has closed yet. A separate workflow checks every 6 hours whether the newest snapshot is over 10 hours old and fails if it is. The 10-hour threshold clears the real 4.6-hour median, so a couple of skipped runs don't trip it while a true outage still does.

2. Platform and API differences

- Polymarket's **/markets** and **/events** were sunset on 2026-05-01 but still return HTTP 200, with the deprecation only in the response headers. Easy to build on a dead endpoint without noticing. I moved to the cursor-paginated **/markets/keyset** before writing any ingestion code.
- Open interest isn't symmetric. Kalshi reports it per market; Polymarket only per event, summed across sibling markets. Assigning that total to one market would misrepresent it, so Polymarket's **open_interest** stays NULL.
- Polymarket's **outcomePrices** is the bid/ask midpoint, not the last trade. I checked it against **bestBid** and **bestAsk** on the same response. A midpoint doesn't go stale on a quiet market the way a last-trade price does.
- Polymarket slugs aren't stable the way Kalshi tickers are: a resolved threshold gets relisted under a new slug. I found three different "Bitcoin dip to \$60k" slugs for the same question, so **external_id** stores Polymarket's numeric id instead.

- Matching strikes don't mean matching contracts. Kalshi's crypto markets ask what BTC or ETH will be worth on one date; Polymarket's ask whether it ever touches a price first. Same asset, similar dollar figure, different question, and only visible in the contract terms. Section 4.1 shows the difference is measurable.

3. Research synthesis

Seven theses logged from three X accounts, recorded secondhand in the `kol_theses` table.

3.1 Resolution risk as an unpriced cost

@thenarrator argues a YES price near \$1 still carries a hidden cost: capital is locked until settlement, so a near-certain bet is capital-inefficient if resolution drags or the exit is thin. He wants a visible carry metric, expected time-to-redemption shown beside price, like a funding rate on perpetual futures. The last few cents, in his view, pay for resolution risk the trader hasn't priced.

@Domahhhh supplies four cases where that risk landed. Polymarket resolved a Venezuela invasion market against its own stated criteria after Trump claimed the US controlled the country, and an Epstein blackmail market YES off one ambiguous email, on a looser standard than it applied days later. Kalshi paid \$0 on a televised Biden and Trudeau meeting because its rules recognized only two sources, neither of which reported it, and settled an Oscars-viewership market on preliminary Nielsen numbers hours before the finals flipped the outcome.

3.2 Trading methodology: holding against consensus

@Domahhhh's Fed Chair trade shows the same read at size. He held a large net-short against the market-implied favorite, who peaked at 85% consensus, for roughly five months and four frontrunner swaps, on his own view that the favorite was a poor fit for monetary policy. The eventual pick matched his lower-probability estimate, netting \$425,000.

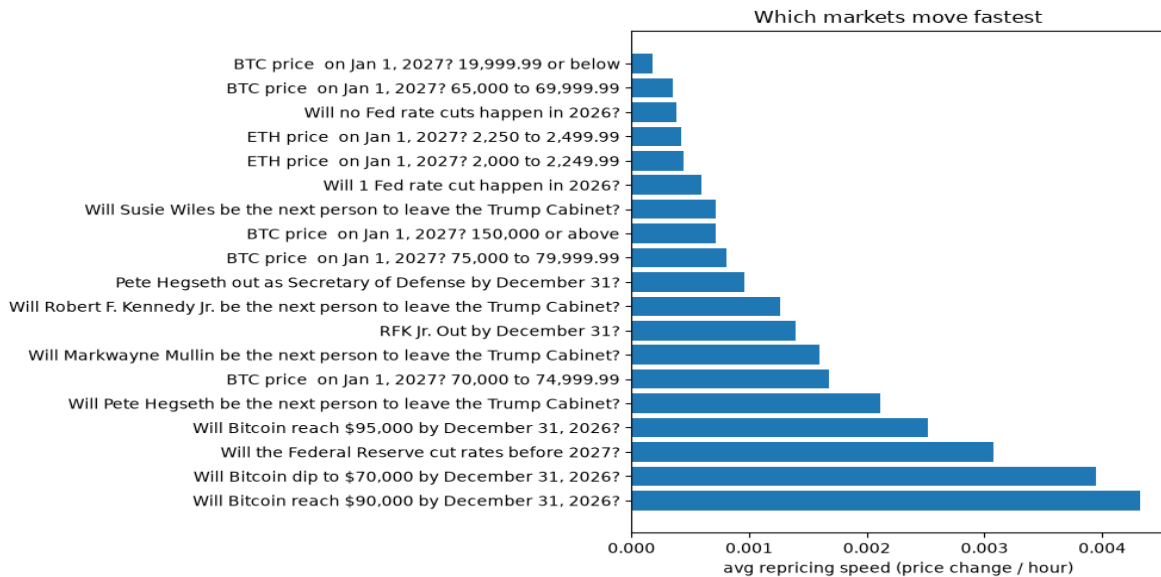
3.3 Theory: why bounded markets discipline claims

@VitalikButerin argues prediction markets are epistemically healthier than social media or unbounded asset markets. A bad take on social media earns unaccountable clout; a bad bet loses real money. Bounded $[0,1]$ pricing also resists the reflexivity and pump-and-dump dynamics unbounded markets are prone to.

4. Findings from the collected data

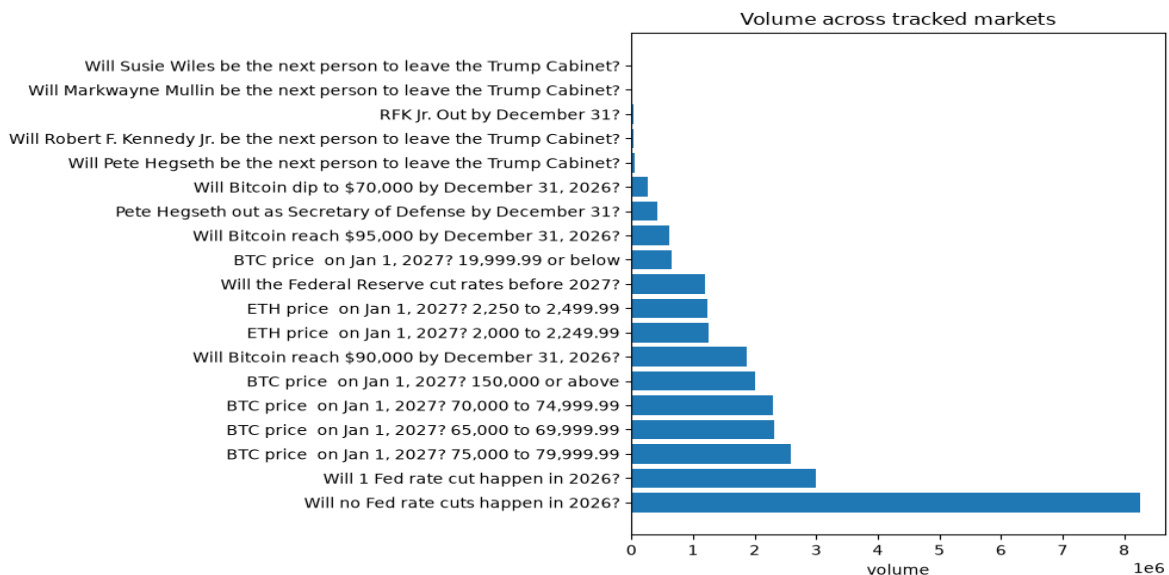
4.1 Repricing speed

Narrator's "prediction-market VIX" made measurable: mean absolute price change per hour, per market, across the window.



I expected the split to fall along crypto against politics. It falls along contract structure. Fastest: "Will Bitcoin reach \$90,000 by December 31, 2026?" at 0.00432 per hour, "Will Bitcoin dip to \$70,000 by December 31, 2026?" at 0.00395, "Will the Federal Reserve cut rates before 2027?" at 0.00308. Slowest: "BTC price on Jan 1, 2027? 19,999.99 or below" at 0.00018, "BTC price on Jan 1, 2027? 65,000 to 69,999.99" at 0.00035, "Will no Fed rate cuts happen in 2026?" at 0.00038. Most of the slow end is Kalshi settle-on-one-date buckets, plus the near-consensus Polymarket rate-cut market, the heaviest book tracked. Bitcoin sits at both ends. A touch-by-deadline contract reprices on every move toward the barrier; a settle-on-one-date bucket only cares where price finishes.

4.2 Volume distribution

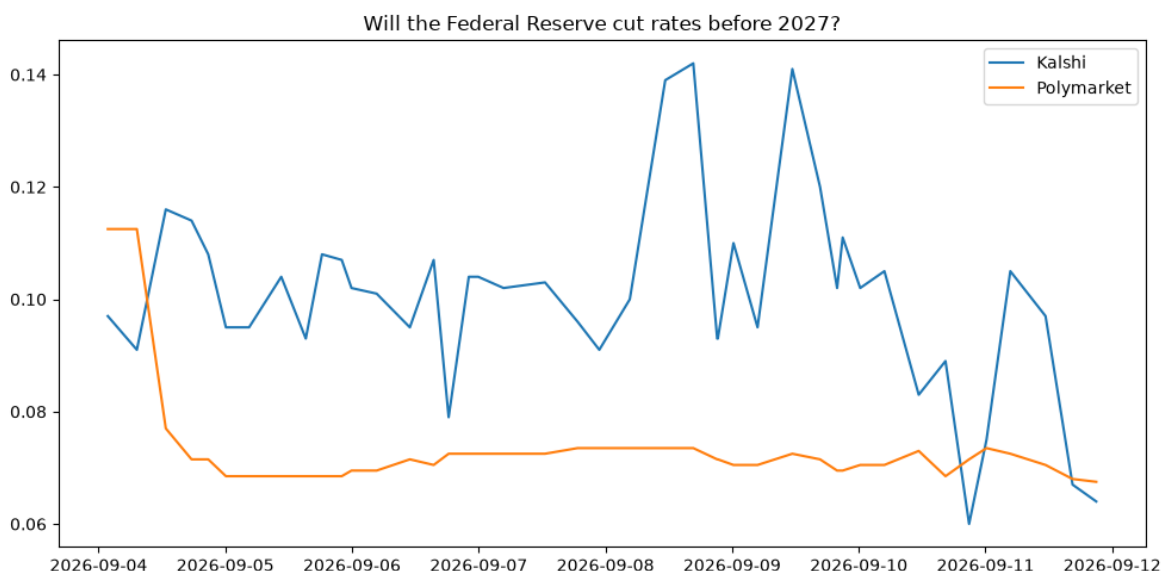


The Polymarket "Will no Fed rate cuts happen in 2026?" market carries about \$8.25M against \$18,148 and \$14,114 on the thinnest Kalshi cabinet markets. It also sits 17th of 19 on repricing speed, so the heaviest book is among the slowest movers without being the slowest. Evercore ISI reportedly found, across five years of settled Kalshi and Polymarket markets, that high-volume markets price more reliably than thin ones. I could not verify that study, so treat it as context rather than evidence from this dataset.

4.3 Cross-platform divergence

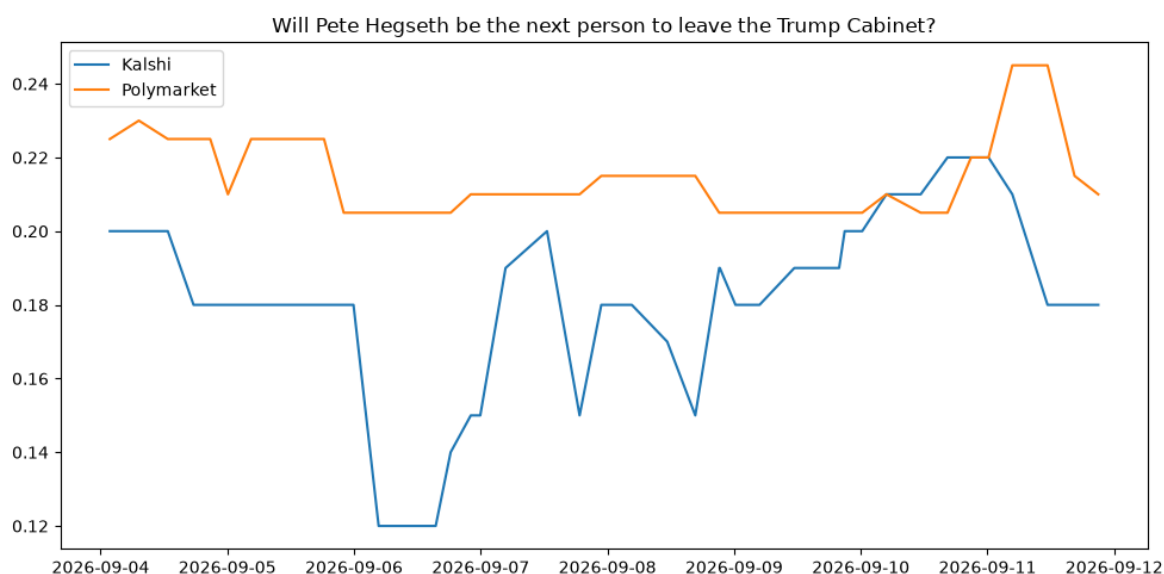
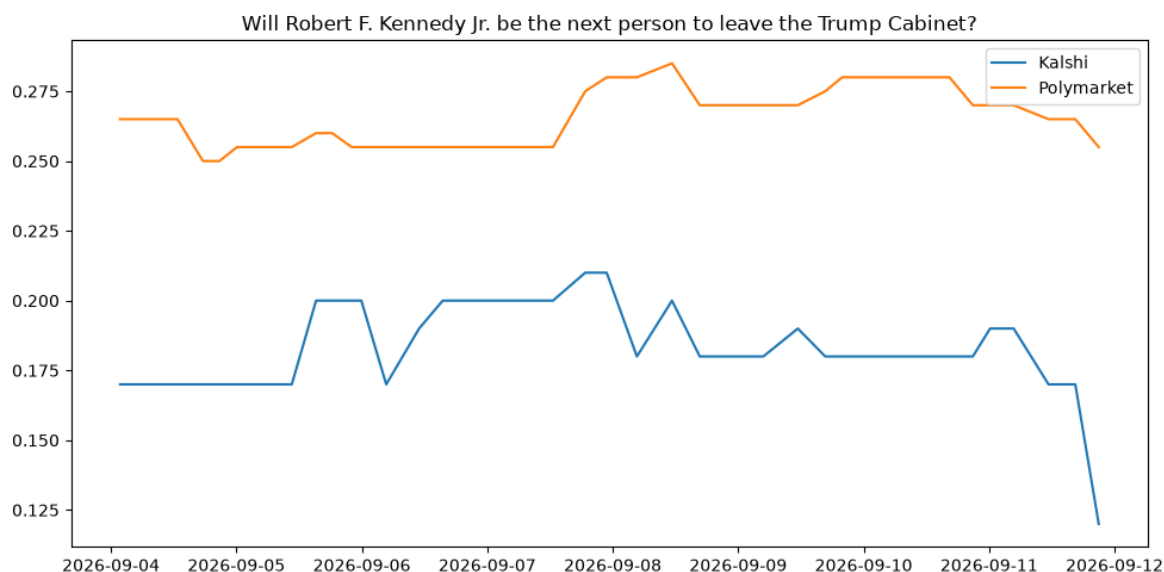
Three of five intended pairs are populated: one macro, two politics. The crypto pairs were never added. The tracked Kalshi crypto markets ask what BTC or ETH will be worth on a specific date while the Polymarket ones ask whether that price is touched before a deadline, so linking them would produce a divergence number that measures nothing.

4.3.1 Macro pair: Fed rate cut



Kalshi's "Will the Federal Reserve cut rates before 2027?" against Polymarket's "Will no Fed rate cuts happen in 2026?", with Polymarket inverted. The inversion matters: the two questions are logically opposite, so a raw comparison showed a false 80-percentage-point divergence. Corrected, the series open on 2026-09-04 at 0.097 and 0.113, separate mid-window, then close on 2026-09-11 at 0.064 and 0.068, a gap of 0.0035. Both drifted down, Kalshi ending marginally below after spiking to 0.142 and giving it back. Across 43 aligned snapshots the ratio averaged 1.38 (median 1.42, standard deviation 0.26, range 0.81 to 1.94). The gap opens wide and closes inside one window rather than holding, which argues for reading it as movement rather than a standing platform difference.

4.3.2 Politics pairs: RFK Jr. and Hegseth



The two politics pairs don't share one pattern. RFK Jr. is tight: mean 0.69, median 0.67, standard deviation 0.07 across 43 points, ranging 0.47 to 0.78, with Kalshi consistently below Polymarket. Hegseth is higher and noisier: mean 0.85, median 0.86, standard deviation 0.12, ranging 0.59 to 1.07. That upper bound clears 1.0, so Kalshi at times priced Hegseth's departure above Polymarket. One structural story about Kalshi's "next to leave" framing sitting below Polymarket's independent yes/no fits RFK Jr. and not Hegseth.

5. Limitations

- 19 markets, a deliberately small fixed watchlist rather than either full catalog.
- 3 of 5 planned links populated. The 2 crypto pairs were dropped once the contract types turned out to differ (price-on-a-date versus touch-by-deadline).

- 9.5 days of observation (2026-09-02 to 2026-09-11), less for the pairs, which start 2026-09-04. The Fed divergence is measured and real, but it closes again by the end, so the mid-window gap isn't a durable platform difference.
- The politics pairs are approximate. Kalshi's cabinet tickers are named KXCABOUT-26MAY22 but close 2029-01-20, the end of the presidential term, against 2026-12-31 on Polymarket. Reading the date out of the ticker gives 2026-05-22, wrong by more than two and a half years, so take it from close_time.
- Polymarket open_interest is unavailable per market (Section 2), so it's NULL throughout.
- Snapshot rows don't record their trigger, so scheduled, manual and local runs are indistinguishable in the delivery figures in Section 1.3.

6. Next steps

- Record a trigger source on every snapshot row so the delivery rate measures scheduled runs alone.
- Track a crypto pair defined the same way on both platforms, which means finding a Kalshi touch-by-deadline market. Section 4.1 suggests the two contract types behave differently enough that pairing them would measure the contract rather than the platform.
- Trade-level ingestion and wallet-level "smart money" tracking via Polymarket's subgraph, scoped out of this week as a stretch goal.
- Widen the watchlist and extend the window to test whether the Fed convergence holds.